Inho Ryu

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Education

University of California, Berkeley – BA in Computer Science — Expected May 2029

Experience

Quantum Physics Intern, Los Alamos National Laboratory – Los Alamos, NM — Apr 2024 – Present

- Conducted research in Quantum group on utilizing gravity sensors for mapping underground through Field Programmable Fate Array chip programming to generate control signals.
- Engaged in 3D CAD design, where I designed, implemented, and tested digital models of quantum sensors to help develop prototypes that would improve performance in detecting various underground objects.
- Developed animations and technical illustrations for a university and lab presentation, effectively communicating gravity sensor applications in subsurface mapping to a multidisciplinary scientific audience.

Astrophysics Researcher, New Mexico State University – Las Cruces, NM — May 2024 – Aug 2024

- Developed a Python library to iteratively compute orbital elements and determine the trajectory of near-Earth asteroid 1660 Wood, achieving <1% error relative to NASA predictions.
- Operated research-grade telescopes at Las Cumbres Observatory and Tortugas Mountain Observatory, applying the Gaussian method with least-squares plate reduction and photometric analysis.
- Created animated orbit visualizations and simulations with Visual Python, supporting collision probability assessments with Earth.
- Co-authored, presented, and submitted a research paper to the IAU Minor Planet Center, including original data contributions, results, and analysis.

Academic Tutor, Breakthrough Collaborative – Santa Fe, NM — Sept 2024 – June 2025

- Designed and led weekly STEM lessons for 15+ first-generation, low-income students, incorporating parent-teacher conferences, individualized feedback, and college counseling to raise average GPA by 0.5.
- Delivered SAT math instruction to students across 4+ countries, improving average scores by 150 points through weekly slide-based lessons and customized practice exams with feedback.
- Facilitated history seminars and tests for 50+ high school students, fostering peer dialogue and critical thinking.
- Tutored 30+ chemistry students, including 10+ in private sessions, raising their average GPA by 0.33.
- Organized enrichment programs and recruited future tutors through school outreach to sustain the program.

Theoretical Computer Science Researcher, Euler Circle – Mountain View, CA — May 2023 – Aug 2023

- Conducted advanced mathematical research with Stanford professors, authoring two 20-page papers—one on the Miller-Rabin, Lucas-Lehmer, and AKS primality tests, and another on Euler's work on continued fractions.
- Submitted the papers for review and presented findings to academic audiences at school and a university-led math circle.

Supercomputing Researcher, Supercomputing Challenge – Santa Fe, NM — Aug 2021 – June 2025

- Researched real-time pandemic monitoring by analyzing COVID-19 YouTube engagement metrics (views, likes, comments) using Python and the YouTube API.
- Applied statistical correlation, lag analysis, and validation with Johns Hopkins case data to demonstrate social media as a viable early-warning system for disease spread.
- Produced technical reports and visualizations for the Supercomputing Challenge.

Skills

Languages: Python, C++, MATLAB
Software: Fusion 360, Inkscape, AutoCAD